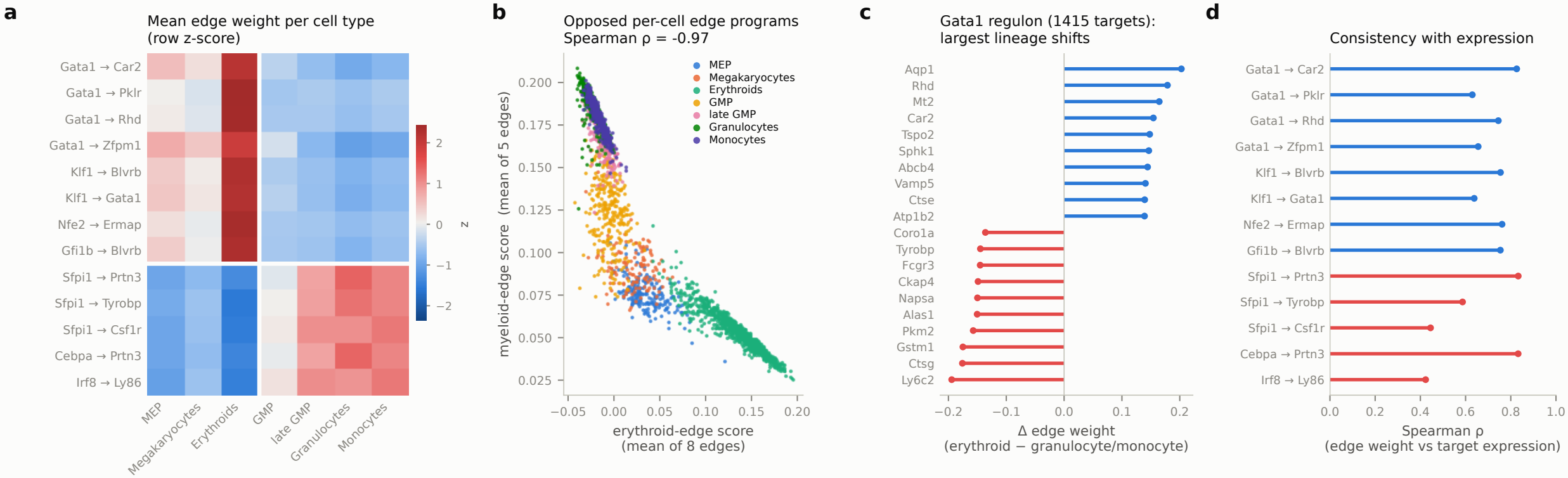
a
Mean edge weight per cell type
(row z-score)
Gata1 → Car2
Gata1 → Pklr
Gata1 → Rhd
Gata1 → Zfpm1
Klf1 → Blvrb
Klf1 → Gata1
Nfe2 → Ermap
Gfi1b → Blvrb
Sfpi1 → Prtn3
Sfpi1 → Tyrobp
Sfpi1 → Csf1r
Cebpa → Prtn3
Irf8 → Ly86
MEP
Megakaryocytes
Erythroids
GMP
late GMP
Granulocytes
Monocytes
z
2
1
0
−1
−2
b
Opposed per-cell edge programs
Spearman ρ = -0.97
MEP
Megakaryocytes
Erythroids
GMP
late GMP
Granulocytes
Monocytes
myeloid-edge score (mean of 5 edges)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
−0.05
0.00
0.05
0.10
0.15
0.20
erythroid-edge score
(mean of 8 edges)
c
Gata1 regulon (1415 targets):
largest lineage shifts
Aqp1
Rhd
Mt2
Car2
Tspo2
Sphk1
Abcb4
Vamp5
Ctse
Atp1b2
Coro1a
Tyrobp
Fcgr3
Ckap4
Napsa
Alas1
Pkm2
Gstm1
Ctsg
Ly6c2
−0.2
−0.1
0.0
0.1
0.2
Δ edge weight
(erythroid − granulocyte/monocyte)
d
Consistency with expression
Gata1 → Car2
Gata1 → Pklr
Gata1 → Rhd
Gata1 → Zfpm1
Klf1 → Blvrb
Klf1 → Gata1
Nfe2 → Ermap
Gfi1b → Blvrb
Sfpi1 → Prtn3
Sfpi1 → Tyrobp
Sfpi1 → Csf1r
Cebpa → Prtn3
Irf8 → Ly86
0.0
0.2
0.4
0.6
0.8
1.0
Spearman ρ
(edge weight vs target expression)